BA ZC581 :
Applied Data Science
Module: 6

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Disclaimer

- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

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Module 6: Data Cleaning & Type Conversion

Module 6 Topics

Topic 1: Handling Missing Data

Topic 2: Type Conversion

Topic 3: Duplicate Removal

Topic 4: Parsing Complex Fields

Topic 5: Demos

Ref: T1 , Ch 5

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Topic 1: Handling Missing Data

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Introduction to Data Pre Processing

Need for Data Preprocessing

Data that gets ingested for analysis comes from multiple heterogeneous sources which may be in a raw format, and differs in structure, format and data types. It could be noisy, incomplete and inconsistent.

Hence, we need to preprocess the data using some transformation operations.

This will enhance the quality of data, reduces data errors and sets up a structured representation of data.

This is an important step in Data Analytics and Machine Learning.

Data preprocessing is also called data cleansing, data cleaning, data wrangling, data scrubbing etc

Data Preprocessing

Improves Data Quality

Raw data often contains missing values, duplicates, or inconsistencies. Preprocessing ensures the dataset is clean and reliable for analysis.

Enhances Accuracy of Models

Well-prepared data reduces noise and bias, leading to more accurate predictions and insights from statistical or machine learning models.

Ensures Consistency Across Sources

When data comes from multiple systems, preprocessing standardizes formats (e.g., dates, units, categories) so they can be meaningfully combined.

Optimizes Performance

By scaling, normalizing, or reducing dimensionality, preprocessing makes algorithms run faster and more efficiently, especially on large datasets.

Core steps in Data Preprocessing

Data Cleaning	• cleaning by filling missing values, smoothening noisy data, removing inconsistencies
Data integration	• Data with different representations are converted to a standard representation, harmonizing data
Data Transformation	• Data is normalized, aggregated, leading to a homogeneous data
Data Reduction	• Generate a reduced representation of data to make storage and computations efficient

Typical Transformations

- Column operations
- Row Operations
- Eliminate / Filter out unwanted data
- Change form and format of data
- Do a range check on data (provided we know the details of the domain)
- Handle missing data (Null values)
- Normalise Data
- Harmonize data
- Homogenize data
- Aggregate data
- Group Data

Data Preprocessing Benefits

Key Benefits

- Data Cleaning: Removes duplicates, handles missing values, and corrects errors.
- Noise Reduction: Smooths irrelevant or incorrect data using techniques like binning, regression, or clustering.
- Data Integration: Merges datasets from multiple sources into a unified view.
- Transformation & Normalization: Converts data into consistent formats and scales values for fair comparison.
- Data Reduction: Minimizes redundancy and reduces dimensionality, making analysis faster and more efficient.

Impact on Analytics

- Accuracy: Clean data ensures models detect real patterns rather than noise.
- Efficiency: Reduced and normalized data speeds up computation.
- Consistency: Integrated datasets prevent contradictions across sources.
- Reliability: Preprocessing builds confidence in insights, supporting strategic decisions.

Example

Imagine analyzing customer purchase data:

Raw dataset: Missing ages, duplicate entries, inconsistent date formats.

After preprocessing:

Missing ages imputed with averages.

Duplicates removed.

Dates standardized to YYYY-MM-DD.

This clean dataset allows accurate trend analysis and reliable predictions.

Exploratory Data Analysis

- Exploratory Data Analysis
  - Helps us summarize and visualize datasets to detect trends, relationships, and anomalies before applying complex models
  - It reveals missing values, outliers, or inconsistencies that need cleaning to ensure reliable analysis
  - By exploring distributions and correlations, EDA informs which statistical or machine learning techniques are most appropriate

EDA Outcomes

- Data Quality Assessment
  - Detects missing values, duplicates, inconsistencies, outliers etc
- Understanding Distributions
  - Summarizes data using statistical operations (mean, median, variance) and plot histograms, box plots to find spread, skewness and central tendency
- Feature Relationships
  - Identifies correlations and dependencies between variables
- Error Detection & Correction
  - Flags incorrect data, error data
- Pattern and Trend Discovery
  - Identifies hidden patterns, clusters or trends
- Model Readiness
  - Identifies most significant features that can be used in the model

EDA Demo

Demo program for EDA

eda_titanic.py

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Missing Data Values

What are they?

Data entries with no recorded value (e.g., blanks, NaN, NULL).	Common in surveys, sensors, logs, or manual entry systems.	Transmission errors can lead to missing values	Can distort analysis, reduce accuracy, and bias results / bias models.	Affect statistical measures

How to handle Missing Values

1. **Deletion**: Remove rows/columns with too many missing values.
2. **Imputation**: Fill with a constant, default, mean, median, mode, or domain-specific values.
3. **Forward / Backward Fill:** Use previous / next valid entry (time series). Perform n-point window average.
4. **Predictive Models:** Estimate missing values using regression or cluster mean or ML .
5. **Flagging**: Add indicator variables to mark missingness

Challenge: Categorical Data

Categorical data without inherent order is a challenge (it can impact popularity counts)

Common Approaches to handle missing data are:

- **Deletion**:
  - Drop rows with missing categories if the proportion is very small.
- **Mode Imputation**:
  - Replace missing values with the most frequent category.
- **Domain/Business Rules:**
  - Use expert knowledge (e.g., “Unknown,” “Other,” or a default category).
- **Predictive Imputation**:
  - Use classification models (e.g., decision trees, k-NN) to predict the missing category.
- **Indicator Variable:**
  - Add a new category like "Missing" to explicitly flag missingness

Demo

Demo program to handle missing values

Missing_Fillup

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Topic 2: Type Conversions

Type Conversions

- Changing data from one type/format to another (e.g., string → integer).
- Ensures consistency, accuracy, and compatibility across datasets and tools.
- Why They Matter?
  - Raw data often mixes types (e.g., numbers stored as text, dates as strings).
  - Models and analytics require correct types for computation and comparison.
- Common Conversions
  - Numeric Conversion: Strings → integers/floats ("45" → 45).
- Categorical Encoding:
  - Nominal values → codes (S,M,L,XL) or one-hot vectors ("Red" → [1,0,0]).
- Date/Time Parsing:
  - Strings → datetime objects ("2025-12-04" → datetime).
- Boolean Conversion:
  - Flags → True/False ("Yes" → True).
- Standardization:
  - Units or formats aligned (e.g., currency, measurement systems).

Encoding

- Transforms data into a machine readable format.
- Converts categorical / text values into numeric or structured representation form
- Enables ML algorithms to process categorical features
- Preserves meaning while ensuring computational compatibility

Common techniques

Label Encoding: Assigns integer codes to categories (Red=0, Blue=1, Green=2).

One-Hot Encoding: Creates binary columns for each category (Red=[1,0,0], Green=[0,1,0], Blue[0,0,1]).

Ordinal Encoding: Maps ordered categories to integers (Low=1, Medium=2, High=3) or (Manager = 9, Officer = 6, Supervisor=4, Assistant = 2).

Frequency/Target Encoding: Replaces categories with statistical values (counts, mean target).

Encoding Demo

Demo program to encode data

Encoding-demo.py

Type-conversions-demo.py

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Topic 3: Duplicate Removal

Duplicate Removal

- Duplicates are data records that appear more than once in the data set.
- Common in merged datasets, logs or manual entries.
- Needs handling as it inflates counts, and skews statistics
- Wastes storage
- Distorts model and its accuracy

Common Techniques to handle duplicates

- Detection: Identify duplicates using unique keys or full row comparison.
- Removal: Drop duplicate rows with tools like pandas.drop_duplicates().
- Retention Rules: Keep first/last occurrence or based on business logic.
- Validation: Ensure critical identifiers (IDs, timestamps) remain consistent.

Duplicate removal demo

Demo program for duplicate removal

Duplicate_demo

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Topic 4: Parsing Complex Types

Complex Data types

What Are Complex Types?

- Data structures beyond simple numbers or strings.
- Examples: JSON, XML, nested lists, hierarchical tables, mixed formats.

Why It Matters?

- Raw data often arrives in semi-structured or nested formats.
- Analytics tools require flattening or normalization for consistency.
- Misinterpreted parsing leads to data loss or incorrect insights.

Parsing Techniques for Complex Data Types

- JSON/XML Parsing: Convert nested objects into tabular form.
- String Tokenization: Split composite fields (e.g., "A,B,C" → separate columns) or a complex EPC is split to constituents.
- Flattening Nested Structures: Expand lists/dictionaries into rows/columns.
- Schema Mapping: Align complex fields with relational database schemas.
- Custom Parsers: Handle domain-specific formats (logs, IoT streams, ASCII streams).
- Regex for complex data structures

Complex Data Parsing Demo

Demo program for parsing complex data

Complex_data_parsing_demo

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Feature Engineering

Feature Engineering

- Features in the data can be made effective
  - Good features = better predictive power.
  - Reduces noise, improves accuracy, and simplifies models.
  - Often more impactful than algorithm choice
  - Need a formal approach to handle features
- Feature Engineering
- The process of creating, transforming, or selecting features (variables) to improve model performance.
- Bridges raw data and machine learning algorithm

Techniques Employed

- Transformation: Scaling, normalization, log transforms.
- Encoding: Converting categorical data (label, one-hot, target encoding).
- Creation: Deriving new features (ratios, interactions, time-based features).
- Reduction: Feature selection, dimensionality reduction (PCA)

Demo Programs

Feature Engineering & PCA

Feature_engineering_demo.py

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Topic 5: Demos of Operations

Summary

- Discussed significance of Data Preprocessing
- Understood EDA operations
- Understood various Pre processing operations like Missing value handling, Type handling, Duplicate handling etc
- Understood how to use parsing techniques
- Studied significance of Feature Engineering

Questions?